

# Zakia Sultana

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## SUMMARY

PhD researcher in EECS at the University of Arkansas seeking a **Research Engineer / R&D internship role in AI-driven EDA and physical design**, where I can apply deep learning to make chip design closure faster and more predictable. I bring hands-on place-and-route experience from industry (Cadence Innovus, Calibre) together with published research on ML-based IR-drop and congestion prediction, including a Best Paper Award at IEEE ICCIT 2025.

## EDUCATION

### University of Arkansas

*Doctor of Philosophy (PhD), Electrical Engineering and Computer Science (EECS)*

**Fayetteville, AR**

*2026 – Expected 2031*

### Chittagong University of Engineering & Technology (CUET)

*B.Sc., Electronics and Telecommunication Engineering | CGPA: 3.66 / 4.00*

**Chattogram, Bangladesh**

*2020 - 2025*

## EXPERIENCE

### Ulkasemi Pvt. Ltd.

*Assistant Engineer, Physical IC Design (PnR)*

**Dhaka, Bangladesh**

*Jan 2026 - Apr 2026*

*Trainee Engineer, Physical IC Design (PnR)*

*Jul 2025 - Dec 2025*

- Delivered PnR blocks at 2nm, 5nm, 14nm, 90nm, and 180nm nodes, meeting timing, DRC, and LVS signoff targets, by running full Cadence Innovus and Synopsys ICC II flows: floorplanning, power planning, placement, clock tree synthesis, and routing.
- Reduced physical verification violations before handoff by analyzing Calibre DRC/LVS reports and applying targeted routing and placement ECO fixes.
- Cut turnaround per PnR iteration by writing TCL scripts to automate repetitive Innovus setup, reporting, and QoR checks in a Linux environment.
- Promoted from Trainee to Assistant Engineer within 6 months by completing structured PnR methodology training and taking ownership of production design tasks.

## SELECTED PUBLICATIONS

- Multiscale Residual Attention UNet+ for Early IR-Drop Prediction and Reliability Optimization** - 28th IEEE ICCIT 2025, **Best Paper Award**. Built a deep learning model that flags voltage-drop hotspots early in the flow, so designers can fix power-grid issues before costly late-stage signoff.
- Proactive VLSI Congestion Prediction Using a Hybrid Residual U-Net with Squeeze-and-Excitation Blocks** - 28th IEEE ICCIT 2025. Designed a lightweight congestion predictor to reduce the number of place-and-route iterations needed for routability closure.
- Spectrum Sensing Using Hybrid Model with Attention Mechanism for Low SNR Environments** - 27th IEEE ICCIT 2024. Improved detection accuracy for signals at -20 dB SNR using an attention-based hybrid model.
- 4 additional IEEE conference papers in intrusion detection, NLP, and computer vision; full list available on request.
- 11 additional IEEE conference papers under review on AI integration in VLSI design.

## SELECTED PROJECTS

- FPGA-Based Real-Time Car Detection** (Python, Verilog, Quartus): Trained a CNN on vehicle image datasets and deployed it on an FPGA with a live camera interface for real-time vehicle-type detection.
- CMOS Logic Cell Design** (Cadence Virtuoso): Implemented schematics and layouts for standard logic gates, adder, decoder, and MUX circuits.

## SKILLS

**EDA / VLSI:** Cadence Innovus, Voltus, Virtuoso, Calibre, Synopsys PrimeTime, IC Compiler II, Quartus, ModelSim.

**Programming:** Python, TCL, C/C++, VHDL, Verilog, MATLAB, Linux, LLM, Agentic AI.

**Machine Learning:** Deep learning (CNN, U-Net, attention, BERT, BiLSTM), scikit-learn models, SHAP, LIME.

**Hardware:** FPGA, Arduino, ESP32, microcontrollers.

## CERTIFICATIONS

- PnR Training (Cadence Innovus and Calibre), Ulkasemi
- IBM Data Science Professional Certificate